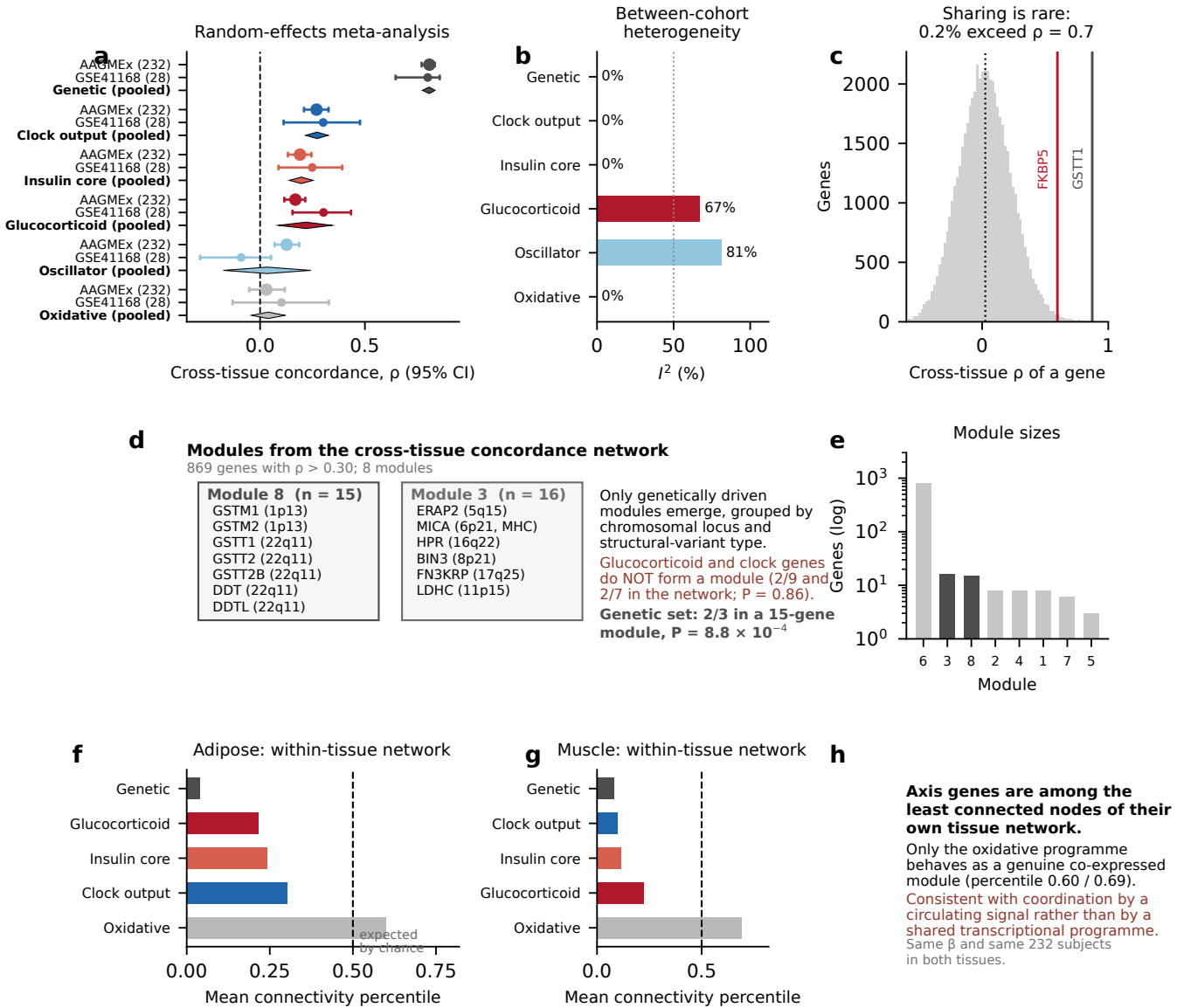
a
Random-effects meta-analysis
AAGMEx (232)
GSE41168 (28)
Genetic (pooled)
Clock output (pooled)
Insulin core (pooled)
Glucocorticoid (pooled)
Oscillator (pooled)
Oxidative (pooled)
0.0
0.5
Cross-tissue concordance, ρ (95% CI)
b
Between-cohort heterogeneity
Genetic
Clock output
Insulin core
Glucocorticoid
Oscillator
Oxidative
0%
0%
0%
67%
81%
0%
0
50
100
I² (%)
c
Sharing is rare:
0.2% exceed ρ = 0.7
Genes
2000
1500
1000
500
0
FKBP5
GSTT1
0
1
Cross-tissue ρ of a gene
d
Modules from the cross-tissue concordance network
869 genes with ρ > 0.30; 8 modules
Module 8 (n = 15)
GSTM1 (1p13)
GSTM2 (1p13)
GSTT1 (22q11)
GSTT2 (22q11)
GSTT2B (22q11)
DDT (22q11)
DDTL (22q11)
Module 3 (n = 16)
ERAP2 (5q15)
MICA (6p21, MHC)
HPR (16q22)
BIN3 (8p21)
FN3KRP (17q25)
LDHC (11p15)
Only genetically driven modules emerge, grouped by chromosomal locus and structural-variant type.
Glucocorticoid and clock genes do NOT form a module (2/9 and 2/7 in the network; P = 0.86).
Genetic set: 2/3 in a 15-gene module, P = 8.8 × 10⁻⁴
e
Module sizes
Genes (log)
10³
10²
10¹
10⁰
6
3
8
2
4
1
7
5
Module
f
Adipose: within-tissue network
Genetic
Glucocorticoid
Insulin core
Clock output
Oxidative
expected by chance
0.00
0.25
0.50
0.75
Mean connectivity percentile
g
Muscle: within-tissue network
Genetic
Clock output
Insulin core
Glucocorticoid
Oxidative
0.0
0.5
Mean connectivity percentile
h
Axis genes are among the least connected nodes of their own tissue network.
Only the oxidative programme behaves as a genuine co-expressed module (percentile 0.60 / 0.69).
Consistent with coordination by a circulating signal rather than by a shared transcriptional programme.
Same β and same 232 subjects in both tissues.